

Analyzing students dropout using penalized logistic regression and hierarchical clustering: a case study in systems engineering

Abstract: Student dropout is a major challenge in higher education, particularly in computer science and systems-related degrees, where attrition rates remain high. This study focuses on the Systems Engineering degree at the Faculty of Exact Sciences (UNICEN) and analyses cohorts from 2011 to 2023 using educational data mining techniques. Academic variables were built from the University database, and student dropout status was defined using objective criteria of inactivity and academic performance. Penalized logistic regression (LASSO) was used to identify the most relevant predictors, while hierarchical clustering, validated using the Silhouette index, was used to segment student profiles. The clustering analysis revealed heterogeneous profiles, ranging from students with strong persistence to those with chronic or early dropout. Findings in this paper confirm the usefulness of data mining for designing early-warning systems and institutional strategies to reduce dropout rates.

Keywords: Data processing, logistic regression, student dropout, academic risk factors

1. Introduction

Recently, higher education programs in software engineering, computer science, and systems development have experienced increased enrollment. This is likely driven by the widespread digitalization and automation of several processes in the industry, as well as the high public visibility of generative Artificial Intelligence (AI) shaping the narrative around future careers [1]. In addition, technology offers good job prospects with attractive remuneration [2]. However, the phenomenon also shows a high student dropout rate; specifically in Argentina, it intensified starting in 2016, with a higher rate than in other degree programs in the educational system [3].

Dropout is conceptualized as a longitudinal process in which students decide whether to continue or abandon their studies based on their level of integration. This multidimensional process includes not only academic factors, but also social factors -such as bonds, friendships, support, and interaction with teachers and peers that profoundly influence a student's sense of belonging and motivation to persist [4].

This paper aims to identify variables associated with student dropout and to characterize student profiles associated with different levels of dropout risk using data mining techniques. Then, this study analyzes academic and demographic data using penalized logistic regression (LASSO) and hierarchical clustering, based on variables such as exam completion, inactivity, living costs, and early academic failures in foundational subjects. In this context, data-driven approaches are very useful for validating insights and for facilitating the design of a proactive early-intervention system featuring institutional strategies that can effectively reduce students' dropout rates and support student retention. It is important to note that this is our first exploratory attempt to apply the proposed techniques, with no prior work on this approach.

1.1. Related work

Integrated techniques that combine advanced statistical models with the analysis of qualitative characteristics to guide institutional interventions aimed at preventing student dropout have evolved in recent years. Some of these approaches are discussed next.

A validated predictive model in [5] was developed primarily using XGBoost and Decision Trees for early-stage dropout in higher education. The model implied preprocessing, feature engineering, and evaluation. Results confirmed that attendance, re-enrollment history, and academic performance are indicators of risk as presented in the literature and the institutional expectations.

In [6], several algorithms were evaluated using data from students at the beginning and end of the semester and from graduated students at Kansas State University, with 15 variables across 215,183 records. According to their results, the algorithms improved their performance after applying regularization techniques to Logistic and Linear regression models, and as the dataset size increased.

Kuna et al. [7] presented a method that uses the TDIDT algorithm to generate rules for characterizing dropout based on available system information. The results showed that induction algorithms are helpful in determining the factors associated with dropout. They highlighted that the results are directly related to data quality.

In [8], data mining and a virtual classroom intervention were used to detect academic performance and enable early prediction of potential dropout in computing degree programs. In [9], a keyword-based bibliometric analysis across 48 articles was used; the factors included communication, degree usefulness, cognitive gain, university entrance score exam, place of residence, number of siblings, and mathematical qualification, among others.

Other approaches for unsupervised machine learning have been proposed, such as K-means, DBSCAN, and agglomerative hierarchical clustering. These algorithms more effectively enable autonomous identification of complex subpopulations than supervised methods. The segments are then validated using specialized metrics such as Silhouette score, Dunn index, Adjusted Rand Index, or cluster Impurity.

In [10], different clustering algorithms for data structures were compared for unlabeled data. Then, advanced data transformation for preprocessing numerical and categorical information, including feature engineering techniques. The approach was validated by using labeled data. The authors claimed that the novelty of the approach lies in integrating clustering with advanced data transformations. The validated grouping studies characterized student dropout profiles in detail, identifying the intersection between academic and demographic vulnerabilities.

According to [11], the challenge for analytical frameworks lies in harmonizing heterogeneous data, numeric and categorical. To this end, advanced feature engineering techniques, such as interval grouping and categorical encoding were used to optimize the performance of the algorithms. In this context, it was found that while K-means excels in data compactness, DBSCAN offers methodological advantages for isolating anomalous trajectories and handling noise in unbalanced populations.

Regardless of the approach, it is important to consider that the practical implementation of these technological tools faces ethical and operational challenges. These systems require compliance with data privacy regulations, such as the General Data Protection Regulation (GDPR), as well as proper management of inherent algorithmic biases. Furthermore, a human-in-the-loop oversight approach is essential to manage the high volume of false positives, thereby ensuring equitable and effective educational support [5].

2. Preliminary data analysis

2.1. Analysis of data provided in the original dataset

The student enrollment dataset was extracted from the SIU-Guaraní¹ system. The database is structured in multiple tables that record all students, including their courses, final exams, and passed subjects.

The original dataset contained data on the entire enrollment of the Faculty of Exact Sciences because some of the researchers conducting this study are affiliated with this institution. To perform the analysis, students from the Systems Engineering program who enrolled between 2011 and 2023 were selected, with a focus on students from the 2011 and 2022 curricula (plans that are still being taken). It is important to note that the dataset was anonymized to protect students' privacy.

Then, using data mining techniques, the original data were preprocessed to reduce noise and create variables that enable analysis and determination of whether a student dropped out of the program.

The variables used in the student dropout analysis were obtained by linking student data to the corresponding exam data (midterms and finals). Those variables are shown in Table 1.

Table 1. Variable names and corresponding descriptions.

Variable name	Description
days_since_enrollment	Time since enrolling in the program, in days
passed_courses	Number of approved courses
regularized_courses	Number of regularized courses
promoted_courses	Number of courses passed by promotion (without final exam)
failed_courses	Number of failed courses
avg_last_year_grades	It is an average grade of the courses taken by the student in the last year.
courses_taken_last_year	Number of subjects enrolled in the last year
avg_perc_completed_courses	Average percentage of courses completed
passed_finals	Number of approved final exams
failed_finals	Number of failed final exams
days_since_last_final	Days since the last final exam was taken
completed_courses	Total number of subjects completed (via final exam or promotion)
avg_perc_finals_taken	Average percentage of final exams taken

2.2 Analysis of the dropped_out variable

For this study, student dropout was defined as a binary outcome variable to characterize students who left the program, named as *dropped_out* variable, and is based on four observable indicators:

1. *Inactivity in final exams*: The student has not registered for a final exam for more than 730 days (2 years).
2. *Low percentage of approved finals*: Less than 50% of final exams approved.
3. *Lack of subject enrollment*: absence of enrollment in courses during the last academic year

¹ University management system Guaraní: <https://www.siu.edu.ar/siu-guarani>

4. *Administrative condition*: the institutional record at SIU Guaraní lists the academic status as Dropout (“Abandono”).

These observable indicators provide an objective measure aligned with academic reality. In this way, the `dropped_out` variable accurately identifies students who have dropped out of the program, distinguishing them from those who, despite low performance, remain academically active, based on the following values:

- True (Did dropout): When the student meets at least one of the four observable indicators.
- False (Did not dropout): When the student does not meet the four observable indicators.

3. Material and methods

To analyze student dropout, two statistical techniques and one statistical method were used.

- a) Penalized logistic regression (LASSO): a technique that helps to address correlated predictors, reduce overfitting, and produce a parsimonious and interpretable model, supporting the identification of relevant risk indicators for targeted institutional interventions [12, 13]. This technique was used to identify the variables most strongly associated with student dropout.
- b) Hierarchical clustering: a technique that helps to facilitate the discovery of subgroups with different patterns and aligns with the descriptive/diagnostic objective of the study, with subsequent validation via Silhouette score. Besides, the Ward.D2 linkage with Euclidean distance was adopted because it minimizes within-cluster variance and produces compact, balanced groups. Compared with complete linkage, Ward.D2 yielded better Silhouette scores and more balanced partitions in our data. The resulting clusters were subsequently evaluated using the Silhouette score. Then, it was selected to identify subgroups of students with different academic progress and dropout patterns without requiring the number of clusters to be specified in advance.
- c) Dendrogram cut: The Silhouette score method was used to determine the dendrogram cut. It measures how closely packed points are within the same cluster and how far a cluster is from the others. It was determined that the optimal number of clusters that maximizes the average Silhouette coefficient is 7 (see Fig. 1).

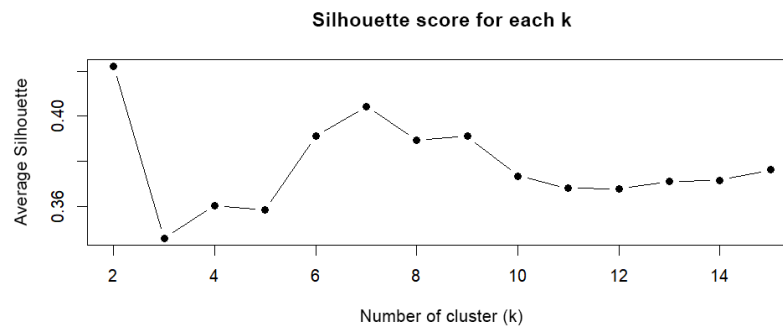


Fig. 1. Calculation of silhouette score for each number of clusters.

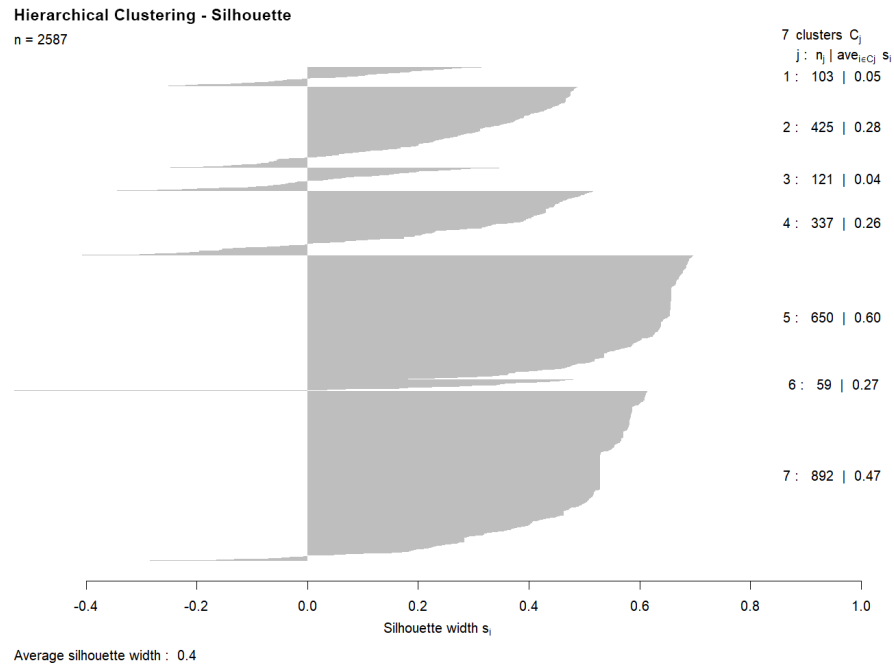


Fig. 2. Silhouette plot for hierarchical clustering performed.

Fig. 2 shows the quality of the 7 clusters obtained by cutting the dendrogram (see Fig. 3) at a height of $h=35$. The average silhouette width is 0.40, which indicates a moderately consistent cluster structure.

It is observed that some groups show good cohesion and separation, such as cluster 5 ($n = 650$, average width ≈ 0.68) and cluster 7 ($n = 892$, average width ≈ 0.47), while others reflect lower grouping quality, such as clusters 1, 2, and 4 (average width ≈ 0.26 - 0.28). The presence of negative values for certain individuals suggests that some points might be assigned to a suboptimal cluster or be close to the boundary between groups.

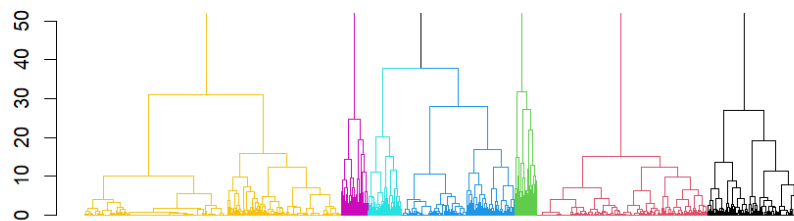


Fig. 3 Dendrogram for hierarchical clustering. Each cluster is differentiated by a different color.

4. Results

4.1 Student dropout analysis

The first analysis performed was exclusively on student dropouts. To perform the analysis, two factors were considered: the student's city of birth and whether they failed any core subject of the program.

4.1.1 Student dropouts by birthplace

Considering that the Systems Engineering program is in Tandil, Buenos Aires, this is an important factor as it can bring additional challenges, such as higher living costs, for example, rent or transportation.

If the student cannot advance in the program, it results in a loss of money and time, which may lead them to drop out, compared with a student born in Tandil.

As Fig. 4 shows, 62.3% of students who were not born in Tandil dropped out.

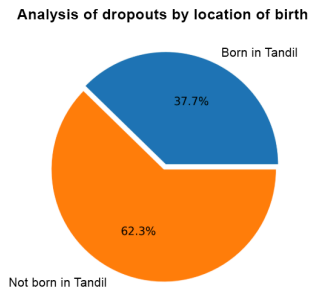


Fig. 4. Pie chart showing the analysis of dropouts by location of birth

4.1.2 Student dropouts due to failure of key courses

Key courses were considered those associated with programming and taught in the first three years. These are Introduction to Programming I and II, Analysis and Design of Algorithms I and II, and Object-Oriented Programming.

These subjects are usually prerequisite courses, and failing any of them results in a one-year delay in the program. Therefore, we observe how many of the students who failed any of these subjects dropped out.

As illustrated in Fig. 5, failing any core subject resulted in a 79.1% dropout rate.

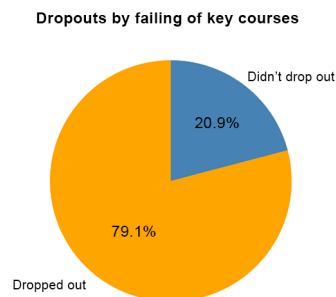


Fig. 5. Pie chart showing the analysis of dropouts due to failure in key courses of the degree program.

4.2 Variable evaluation

To statistically support the factors associated with dropout, we evaluated the influence of the indicators obtained in the preliminary analysis on the dropout variable. Previously, all non-numerical variables and those considered less relevant or redundant to the analysis (such as *regularized_courses* and *completed_courses*) were removed.

4.3 Correlation matrix

To evaluate the consistency between the independent variables and the dependent variable *dropped_out*, an interpretation of the correlation matrix generated prior to hierarchical clustering was first performed (Table 2). In addition, we sought strong correlations among independent variables within the same correlation matrix to identify collinearities, which we then addressed by applying penalized logistic regression (LASSO) (Table 3).

Table 2. Correlation matrix between the independent variables and the dependent variable *dropped_out*.

Variable	Correlation with <i>dropped_out</i>	Interpretation
days_since_last_final	+0.6939	Strong positive correlation: the longer the time since the last final exam, the greater the likelihood of dropping out. Strong association with dropout.
avg_perc_finals_taken	-0.3502	Moderate negative correlation: the greater the progress in the degree program, the lower the probability of dropping out.
passed_finals	-0.3556	Like the above: more final exams passed → lower dropout rates.
failed_courses	+0.3148	Moderate correlation: may suggest frustration → drop out.
passed_courses	-0.2270	Slight negative correlation: more subjects passed → lower dropout rate.
courses_taken_last_year	-0.1830	Not being active recently is associated with higher dropout rates.
days_since_enrollment	+0.4979	The longer the time since admission, the greater the likelihood of dropping out.
failed_finals	-0.1964	Slight negative correlation, possibly indicating persistence.
promoted_courses	+0.0204	There does not appear to be a clear correlation with dropout rates.

Table 3. Strong correlations between the variables taken that could cause collinearity in models.

Variables	Correlation	Interpretation
avg_perc_finals_taken and passed_finals	0.9994	Almost perfect: they are almost the same (they can generate redundancy in models).
passed_courses and passed_finals	0.9095	High correlation: multicollinearity may be present.
passed_courses and avg_perc_finals_taken	0.9095	Very high: these are two variables that measure progress.
days_since_last_final and days_since_enrollment	0.8681	Highly correlated: may represent "recent inactivity" combined with seniority.

4.3.1 Logistic regression – LASSO model

set.seed(1) was used to ensure reproducibility across cross-validation runs. Since lambda is the penalization parameter determined by cross-validation, we adopted lambda.min, the value that minimizes the estimated out-of-sample prediction error, and the coefficients (see Table 4 for coefficient values and interpretations) were extracted at the regularization path point determined by lambda.min. Sample size used was 2587 students.

4.4 Cluster analysis

Based on the averages of academic variables for each cluster, we can create student profiles that characterize everything from active excellence and regular progress to low active progress, near-graduation/graduation, early/mid dropout, or chronic dropout. Clusters 1, 4, and 7 were selected because they are the most representative and informative cases, based on performance and dropout rates across the variables analyzed, yielding characteristic results.

The decision is also supported by the dendrograms shown in Fig. 6, Fig. 7, and Fig. 8. Upon examination, the data points cluster at low heights, indicating a high correlation among them.

Table 4. Coefficients and interpretation of variables associated with the risk of dropout.

Variable	Coefficient	Interpretation
days_since_enrollment	-0.000146	The longer the time since admission, the probability of dropout decreases slightly. Very small effect, candidate for discarding.
passed_courses	-0.0358	Each course passed reduces the chances of dropping out.
promoted_courses	0	It was not selected by the model; it does not provide stable predictive value.
failed_courses	-0.0911	A higher number of failed courses reduces the probability of dropping out. This could be noise, or it could indicate persistence despite failing.
courses_taken_last_year	-2.435	The more activity there was last year, the lower the chance of dropping out. It has a very strong effect with a very high absolute value.
passed_finals	-0.0165	The more finals passed, the lower the chance of dropping out. It has a slight effect.
failed_finals	+0.0084	The more failed exams, the greater the chances of dropping out. Almost no effect.
days_since_last_final	+0.0243	The longer students go without taking a final exam, the greater the likelihood of them dropping out.
avg_perc_finals_taken	-10.238	The higher the percentage of the degree completed, the lower the risk of dropping out. It is the variable with the highest absolute value and therefore the most influential.

4.4.1 Cluster 1

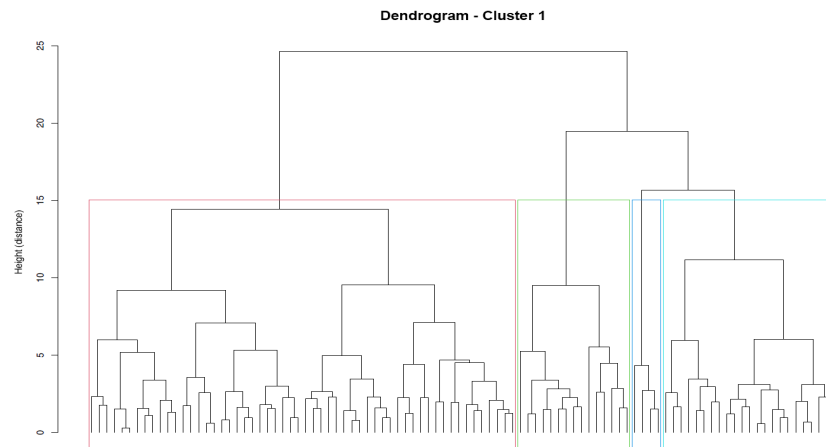


Fig. 6. Dendrogram of cluster 1 (purple dendrogram of Fig. 3)

Student profile: students falling behind and at risk of dropping out.

Variable results: courses passed: 17.8 (high), courses failed: 22.3 (very high), finals passed: 10.3, finals failed: 5.3, promotions: almost none, percentage of finals taken: 28.9%, time since enrollment: 4,339 days (~12 years); dropout rate: 93.8%, last final taken more than 6 years ago.

Cluster description: these are students who have been in the program for a long time, have passed a good number of courses but have also accumulated many failures. Few have advanced and have had several failed attempts at finals. Most ended up dropping out. They require an accelerated completion plan (flexible schedules, retakes, tutoring).

Interpretation: this profile suggests prolonged enrollment; the accumulation of many failed courses and prolonged inactivity may be associated with a higher probability of dropout.

4.4.2 Cluster 4

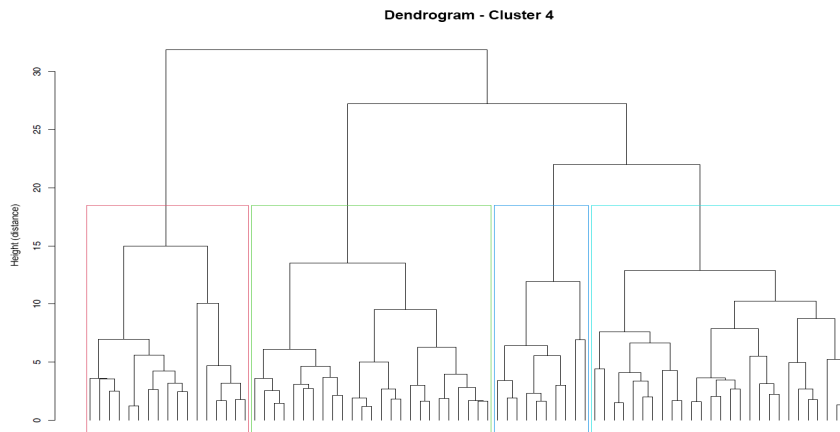


Fig. 7. Dendrogram of cluster 4 (green dendrogram of Fig. 3)

Student profile: advanced students with a solid track record, close to graduation.

Variable results: courses passed: 23.7 (very high), courses failed: 12.7 (moderate), finals passed: 13.8 (high), finals failed: 5.4, promotions: 3.3 (significant), percentage of finals taken: 37% (the highest), time since enrollment: 2,484 days (~7 years), dropout rate: 11.4%, last final taken: ~2 years ago

Cluster description: these are students with a high level of academic progress who have passed many courses and taken numerous final exams, even after several promotions. Although some failed exams, most are still in the program and have not dropped out.

Interpretation: Despite being advanced students with many exams taken and subjects passed, dropouts from this group may have left due to external factors (work, personal issues, career change) or accumulated fatigue from so many years of study (~7 years).

4.4.3 Cluster 7

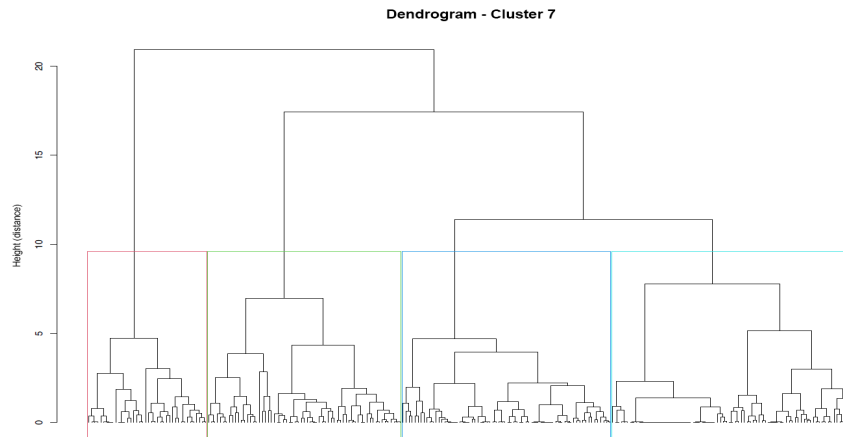


Fig. 8. Dendrogram of cluster 7 (blue dendrogram of Fig. 3)

Student profile: students in regular degree programs, with steady progress and good exam performance.

Variable results: courses passed: 10.1 (moderate-low), courses failed: 1.9 (low), finals passed: 10.1 (very high compared to courses taken), finals failed: 1.9, promotions: practically none, percentage of finals taken: 27%, time since enrollment: 1,442 days (~4 years), dropout rate: 0%, last final taken just over a year ago

Cluster description: They are relatively young students in their degree programs, with good performance in final exams and a low failure rate. Although they don't pass courses through promotion, they take final exams and pass frequently. None have dropped out.

Interpretation: No dropouts detected in this group. Academic performance is stable, with a good pass rate for final exams and a moderate length of stay (~4 years).

4.5 Comparison with past related works

These results can be contrasted with the related work presented in Section 1.1. The predictors selected by LASSO — recent inactivity in course enrollment, the percentage of finals taken, and the time elapsed since the last final exam — are consistent with the risk indicators reported in [5], where attendance, re-enrollment history, and academic performance emerged as the strongest early-stage signals: in both cases, behavioral markers of disengagement outperform purely grade-based measures.

Also, the findings in our study extend [6], which showed that regularization improved the performance of logistic and linear regression models across 215,183 records at Kansas

State University; the present study demonstrates the applicability of L1- penalized logistic regression remains effective on a substantially smaller cohort ($n = 2,587$), where the risk of overfitting under strong collinearity — such as the 0.9994 correlation between `avg_perc_finals_taken` and `passed_finals` reported in Table 3 — is comparatively higher.

With respect to [7], where TDIDT induction rules were used to characterize dropout and the authors emphasized that results depend directly on data quality. The approach presented in this research addresses that limitation by defining `dropped_out` using an explicit, reproducible criterion based on inactivity and academic performance, rather than relying solely on the administrative status recorded in SIU-Guaraní.

Finally, the contextual results align with the bibliometric evidence in [9], which identified place of residence as a factor recurrently associated with dropout: the finding that 62.3% of dropouts were not born in Tandil provides local quantitative support for that factor. The percentage of dropouts who failed at least one core programming subject reinforces the emphasis placed in [8] on early detection within computing degree programs, where foundational subjects operate as prerequisite bottlenecks. In that study, the highest dropout rates occur during the first two years of the program, precisely when the largest number of core programming courses are offered.

5. Conclusions and future works

A major challenge in higher education, particularly in computer science and systems-related degrees, is the high dropout rate in bachelor's programs. The results of the analysis presented in this paper provide a deeper understanding of the factors associated with student dropout rates, with a specific focus on the Systems Engineering program at the Faculty of Exact Sciences (UNICEN). The results show that dropout rates are strongly linked to quantifiable academic variables: (1) the time elapsed since the last final exam taken, (2) the number and percentage of finals passed, and (3) recent inactivity in course enrollment.

Another variable analyzed was the promotion of courses, which was not shown to be a possible factor in dropout rates, because not all courses in the degree program are promotable. Besides, if all of them were promoted, then it would indeed be a variable that would show better progress between one student and another.

Regarding the methods used to perform the analysis, hierarchical clustering proved an effective tool for segmenting students into distinct groups, identifying everything from high-achieving and persistent students to those who drop out early or late in their studies.

The study also revealed the influence of contextual factors:

- 62.3% of dropouts were not originally from the city of Tandil, suggesting that geographical or socioeconomic factors may be associated with dropout. However, socioeconomic variables such as housing costs, employment status, and family support were not directly measured in the current dataset.

- 79.1% of dropouts failed at least one of the core subjects in the first years (Introduction to Programming I and II, Algorithm Analysis and Design I and II, Object-Oriented Programming), which may indicate a strong influencing factor when considering dropping out.

Based on the above, the institution can target specific support policies.

As future work, various lines of continuity can be proposed: (a) data enrichment, including socioeconomic and contextual indicators (such as employment status, family support and motivation); (b) model validation, applying other clustering techniques, such as K-Means, DBSCAN, and alternative supervised models such as Random Forest, Gradient Boosting, or neural networks, (c) institutional deployment, analyzing other

faculties within the same university, or even other institutions, (d) design an institutional tool that, based on the identified indicators, would allow early warnings to be issued about students at risk and preventive interventions to be activated (e.g., scholarships, flexible exam schedules, and academic guidance).

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